

Jason Yang

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Summary

My career objective is to develop the modern artificial intelligence (AI) foundations for biomolecular engineering. Some of my notable contributions include pioneering methods for AI-assisted protein optimization, generative models for protein design, and deep learning methods to discover functional enzymes.

Education

California Institute of Technology

Pasadena, CA

PH.D. IN CHEMICAL ENGINEERING

2021 - 2026

Co-advisor: Frances Arnold (Chemical Engineering, Biochemistry, Biological Engineering)

Co-advisor: Yisong Yue (Computing & Mathematical Sciences)

GPA: 4.0/4.0 | **Thesis:** Artificial Intelligence Methods for Enzyme Engineering

Yale University

New Haven, CT

B.S. IN CHEMICAL ENGINEERING | B.S. IN APPLIED MATH

2017- 2021

GPA: 3.93/4.00

New Trier High School

Winnetka, IL

HIGH SCHOOL DIPLOMA

2013- 2017

GPA: 5.33/5.33

Honors & Awards

- 2024 - 2026** Google PhD Fellowship
- 2021 - 2024** NSF Graduate Research Fellowship
- 2020 - 2021** Barry Goldwater Scholarship
- 2019** Tau Beta Pi Engineering Honor Society
- 2018** First-Year Summer Research Fellowship, Yale University
- 2017** Regeneron Science Talent Search Top Scholar
- 2017** National Merit Scholarship Finalist
- 2017** Congressional Award for Youth Gold Medal
- 2017** Science Olympiad National Medalist

Experience

Graduate Research at Caltech

Pasadena, CA

ADVISORS: PROF FRANCES ARNOLD, PROF YISONG YUE

2021 - present

- Proposed a method using diffusion models for protein generation and optimization
- Created methods for uncertainty quantification and Bayesian optimization with deep kernels for protein optimization
- Developed protein language models for enzyme generation and optimization
- Pioneered multi-modal models for enzyme retrieval and annotation
- Developed graph neural networks for representation learning of physics-informed features
- Performed computational scoring and filtering of enzyme candidates
- Wrote successful proposals for individual fellowships from NSF (\$159k) and Google (\$206k)
- Contributed to grant writing for DOE, NSF, Amgen Awards
- Collaborated with various groups at Caltech, Microsoft, Elegen Bio

Summer PhD Intern at Profluent Bio

Berkeley, CA

SUPERVISORS: DR. JEFFREY RUFFOLO, AADYOT BHATNAGAR, DR. ALI MADANI

2024

- Developed a method for conditional generation of enzymes using protein language models

Undergraduate Research at Yale

New Haven, CT

ADVISORS: PROF MENACHEM ELIMELECH, PROF YING LI, PROF SMITA KRISHNASWAMY

2018 - 2021

- Implemented machine learning models to predict the permeability and selectivity of polymeric gas separation membranes
- Evaluated the parameters controlling water-organic separations in nanoporous graphene membranes using molecular dynamics simulations
- Investigated cation-specific selectivity in membranes by synthesizing polyelectrolytes and incorporating them into novel nanofiltration membranes using layer-by-layer assembly
- Studied the mechanism of ecotoxicology and environmental impact of MnO₂ nanomaterials using dynamic light scattering and experimental assays

Undergraduate DoE National Lab Summer Intern

Golden, CO

NATIONAL RENEWABLE ENERGY LAB (NREL) | SUPERVISOR: DR. HEATHER MAYES

2020

- Implemented machine learning models using features from molecular dynamics simulations to accurately predict mechanical properties of bio-based polymers

Undergraduate Pharma Technical Development Summer Intern

South San Francisco, CA

GENENTECH | SUPERVISOR: STEVE RUSSELL

2019

- Automated an ICP-MS trace-metals assay for greater throughput by developing software and hardware

High School Summer Research Intern

Evanston, IL

NORTHWESTERN | ADVISOR: PROF TERI ODOM

2016

- Developed a cost-effective nanoscale patterning method of polyolefin thin films for enhanced hydrophobicity

Publications

>700 citations and h-index of 10 according to Google scholar. *indicates shared first-authorship

First-Author Papers

1. **Yang, J.***; Chu, W.*; Khalil, D.; Astudillo, R.; Wittmann, B.J.; Arnold, F. H.; Yue, Y. Steering Generative Models with Experimental Data for Protein Fitness Optimization. *Advances in Neural Information Processing Systems* 2025.
2. **Yang, J.***; Lal, R. G.*; Bowden, J. C.; Astudillo, R.; Hameedi, M. A.; Kaur, S.; Hill, M.; Yue, Y.; Arnold, F. H. Active Learning-Assisted Directed Evolution. *Nat. Commun.* 2025, 16 (1), 714.
3. **Yang, J.**; Mora, A.; Liu, S.; Wittmann, B. J.; Anandkumar, A.; Arnold, F. H.; Yue, Y. CARE: A Benchmark Suite for the Classification and Retrieval of Enzymes. *Advances in Neural Information Processing Systems* 2024.
4. **Yang, J.**; Bhatnagar, A.; Ruffolo, J. A.; Madani, A. Function-Guided Conditional Generation Using Protein Language Models with Adapters. *Machine Learning for Structural Biology Workshop at NeurIPS* 2024.
5. **Yang, J.**; Ducharme, J.; Johnston, K. E.; Li, F.-Z.; Yue, Y.; Arnold, F. H. DeCOIL: Optimization of Degenerate Codon Libraries for Machine Learning-Assisted Protein Engineering. *ACS Synth. Biol.* 2023, acssynbio.3c00301.
6. **Yang, J.***; Tao, L.*; He, J.*; McCutcheon, J. R.; Li, Y. Machine Learning Enables Interpretable Discovery of Innovative Polymers for Gas Separation Membranes. *Sci. Adv.* 2022, 8 (29), eabn9545.
7. **Yang, J.**; Shen, Z.; He, J.; Li, Y. Efficient Separation of Small Organic Contaminants in Water Using Functionalized Nanoporous Graphene Membranes: Insights from Molecular Dynamics Simulations. *Journal of Membrane Science* 2021, 630, 119331.

Review/Perspective Papers

8. **Yang, J.**; Li, F.-Z.; Long, Y.; Arnold, F. H. Illuminating the Universe of Enzyme Catalysis in the Era of Artificial Intelligence. *Cell Systems* 2025.
9. **Yang, J.**; Li, F.-Z.; Arnold, F. H. Opportunities and Challenges for Machine Learning-Assisted Enzyme Engineering. *ACS Cent. Sci.* 2024, 10 (2), 226–241.

Co-Authored Papers

10. Lambert, T.; Tavakoli, A.; Dharuman, G.; **Yang, J.**; Kaur, S.; Hill, M.; Ramanathan, A.; Arnold, F. H. Sequence-Based Generative AI-Guided Design of Versatile Tryptophan Synthases. *Under Review, Nat. Comm.*
11. Long, Y.; Mora, A.; Abbasinejad, F.; Li, F.-Z.; Reinprecht, P.; Wittmann, B.; Kennemur, J.; Carder, H.; **Yang, J.**; Lambert, T.; O'Meara, R.; Radtke, L.; Qin, Z.; Brinkmann-Chen, S.; Arnold, F. Enzyme Engineering Database (EnzEngDB): A Platform for Sharing and Interpreting Sequence-Function Relationships across Protein Engineering Campaigns. *Under Review, Nucleic Acids Res.*
12. Yang, K.; **Yang, J.**; Yue, Y.; Arnold, F.H. Classifier-Free Guidance with Diffusion Models for Function-Guided Protein Generation. *Under Review, Machine Learning for Structural Biology Workshop.*
13. Lal, R. G.; **Yang, J.**; Zhang, Z.; Li, F.-Z.; Arnold, F.H. Machine Learning-Assisted Evolution of Promiscuous Enzyme Libraries. *Under Review.*
14. Mora, A.; Reisenbauer, J. C.; Schmid, H.; Miyazaki, I.; Long, Y.; **Yang, J.**; O'Meara, R.; Brinkmann-Chen, S.; Arnold, F. H. Enzyme-Tk: Searching the Sequence Space to Find New Enzymatic Starting Points for Bioremediation. *Under Review.*
15. Li, F.-Z.; **Yang, J.**; Johnston, K. E.; Guersoy, E.; Yue, Y.; Arnold, F. H. Evaluation of Machine Learning-Assisted Directed Evolution Across Diverse Combinatorial Landscapes. *Cell Systems* 2025, 101387.
16. Viswesh, A.; **Yang, J.**; Arnold, F. H.; Yue, Y. Evaluating Multi-Modal Models for Enzyme-Reaction Retrieval. *Workshop on Foundational Models for Life Sciences at ICML 2025.*
17. Johnston, K. E.; Almhjell, P. J.; Watkins-Dulaney, E. J.; Liu, G.; Porter, N. J.; **Yang, J.**; Arnold, F. H. A Combinatorially Complete Epistatic Fitness Landscape in an Enzyme Active Site. *Proc. Natl. Acad. Sci.* 2024, 121 (32), e2400439121.
18. He, J.; **Yang, J.**; McCutcheon, J. R.; Li, Y. Molecular Insights into the Structure-Property Relationships of 3D Printed Polyamide Reverse-Osmosis Membrane for Desalination. *Journal of Membrane Science* 2022, 658, 120731.
19. DuChanois, R. M.; Heiranian, M.; **Yang, J.**; Porter, C. J.; Li, Q.; Zhang, X.; Verduzco, R.; Elimelech, M. Designing Polymeric Membranes with Coordination Chemistry for High-Precision Ion Separations. *Sci. Adv.* 2022, 8 (9), eabm9436.
20. Zucker, I.; Hashmi, S. M.; **Yang, J.**; He, Y.; Pfefferle, L. D.; Elimelech, M. Shape-Dependent Interactions of Manganese Oxide Nanomaterials with Lipid Bilayer Vesicles. *Langmuir* 2019, 35 (43), 13958–13966.

Presentations

May 2026	Protein and Antibody Engineering Summit , Invited Oral Presentation	Boston, MA
Dec 2025	NeurIPS Main Conference , Poster	San Diego, CA
Nov 2025	Caltech-UChicago AI+Science Conference , Poster	Pasadena, CA
Nov 2025	AIChE Annual Meeting , Poster	Boston, MA
Aug 2025	BioML Symposium , Invited Oral Presentation	Prague, CZ
Apr 2025	GEM Workshop at ICLR , Poster	Singapore, SG
Apr 2025	Stanford AI for Science Workshop , Poster	Stanford, CA
Mar 2025	Caltech AI in Biology Symposium , Poster	Pasadena, CA
Dec 2024	NeurIPS Main Conference , Poster	Vancouver, BC
Dec 2024	Machine Learning for Structural Biology Workshop at NeurIPS , Poster	Vancouver, BC
Sep 2024	Debora Marks Lab Reading Group , Invited Oral Presentation	Virtual
Jul 2024	Machine Learning for Protein Engineering Seminar Series , Oral Presentation	Virtual
Jul 2024	Caltech ML for Chemistry Seminar Series , Oral Presentation	Pasadena, CA
Jun 2024	Simons Institute Workshop on AI for Science , Oral Presentation	Berkeley, CA
Oct 2023	Caltech CCE Seminar Day , Oral Presentation Winner	Pasadena, CA

Teaching & Mentorship

Mentorship

CALTECH

- Sonia Yuan – Ph.D. Biochemistry, Caltech (expected 2029)
- Masy Domecillo – Ph.D. Biochemistry, Caltech (expected 2029)
- Zahra Shivji – Ph.D. Chemical Engineering, Caltech (expected 2028)
- Kerui Yang – Undergraduate Computer Science, University of Sydney (expected 2027)
- Daniel Khalil – Undergraduate Computer Science, Caltech (2026)
- Annika Viswesh – Undergraduate Computer Science, Caltech (2026)
- Tanvi Ganapathy – Undergraduate Bioengineering, Caltech (2026)
- Bryan Wang – Undergraduate Chemical Engineering, Caltech (2025), Current Ph.D. at MIT
- Emre Guersoy – M.S. Biotechnology, ETH Zurich (2023)

Pasadena, CA

Spring 2023 - Present

Graduate Teaching Assistant

CALTECH

- BE/ChE163 (Fall 2025): Introduction to Biomolecular Engineering (instructor Steve Mayo)
- BE/ChE163 (Fall 2024): Introduction to Biomolecular Engineering (instructor Steve Mayo)
- ChE165 (Fall 2023): Graduate-Level Chemical Thermodynamics (instructor Zhen-Gang Wang)
- Duties: Delivering selected lectures, developing new problem sets, office hours, grading

Pasadena, CA

Fall 2023 - Present

Undergraduate Teaching Assistant

YALE UNIVERSITY

- CENG 150: Intro to Engineering Analysis (instructor Michael Loewenberg)
- CHEM 333b: Physical Chemistry with Applications in the Physical Sciences II (instructor Patrick Vaccaro)

New Haven, CT

Fall 2019 - Spring 2021

Service & Outreach

Academic & Professional Service

REVIEWING

- ACS Central Science, Briefings in Bioinformatics, ACS Catalysis, ACS Synthetic Biology, MLSB NeurIPS Workshop, GEM ICLR Workshop

ICLR VOLUNTEER CHAIR

2025

- Recruited and assigned duties for volunteers at the International Conference on Learning Representations to promote diversity and inclusion

Leadership & Outreach

CO-ORGANIZER FOR THE ML FOR PROTEIN ENGINEERING SEMINAR SERIES

Fall 2024 - Present

- Invited speakers, scheduled and hosted seminars, and managed social media for a community of >3000 members worldwide

SCIENCE OLYMPIAD VOLUNTEER

Spring 2024 - Present

- Organized science activities for middle school and high school students at the Southern California State Science Olympiad tournament

IGEM MENTOR

Spring 2023 - Present

- Served as a project mentor for high school and undergraduate students participating in the International Genetically Engineered Machine competition